

New Spaces from Old

General Topology

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Remark

In choosing a topology for a subset $S \subseteq X$, there are two competing priorities: we would like the inclusion map $S \hookrightarrow X$ to be continuous (from which it follows by composition that the restriction to S of any continuous map $f : X \rightarrow Y$ is continuous); and we would also like continuous maps into X whose images happen to lie in S also to be continuous as maps into S . For the first requirement, S needs to have enough open subsets, and for the second it should not have too many. The subspace topology is chosen as the optimal compromise between these requirements.

Definition

Let $(X, \mathcal{T})$ be a topological space, and let $S \subseteq X$ be any subset. We define a topology $\mathcal{T}_S$ on S by

$$\mathcal{T}_S = \{U \subseteq S : U = S \cap V \text{ for some } V \in \mathcal{T}\}.$$

The topology $\mathcal{T}_S$ is called the **subspace topology** (or sometimes the **relative topology**) on S . A subset of a topological space X , considered as a topological space with the subspace topology, is called a **subspace of X** .

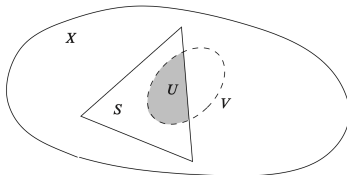


Figure 1: An open subset of the subspace S

Remark

*For clarity, if S is a subspace of X , we sometimes say that a subset $U \subseteq S$ is **relatively open** or **relatively closed in S** , to emphasise that we mean open or closed in $\mathcal{T}_S$, not in $\mathcal{T}$. Similarly, if x is a point in S , a neighbourhood of x in S in the subspace topology is sometimes called a **relative neighbourhood of x** .*

Example 3.1

Consider the subspace $S_1 = [0, 1] \cup (2, 3)$. Show that the interval $[0, 1]$ is not open in $\mathbb{R}$, but it is open in S_1 .

Proposition 3.2

Suppose S is a subspace of X .

- i) If $U \subseteq S \subseteq X$, U is open in S , and S is open in X , then U is open in X . The same is true with “closed” in place of “open”.*
- ii) If U is a subset of S that is either open or closed in X , then it is also open or closed in S , respectively.*

Remark

Natural topologies such as the subspace topology can usually be characterised in terms of which maps are continuous with respect to them. This is where the following theorem acquires its name, and we make this precise with the uniqueness theorem following it.

Theorem 3.3 (Characteristic Property of the Subspace Topology)

Suppose X is a topological space and $S \subseteq X$ is a subspace. For any topological space Y , a map $f : Y \rightarrow S$ is continuous if and only if the composite map $\iota_S \circ f : Y \rightarrow X$ is continuous:

$$\begin{array}{ccc} & & X \\ & \nearrow \iota_S \circ f & \uparrow \iota_S \\ Y & \xrightarrow{f} & S. \end{array}$$

Corollary

If S is a subspace of the topological space X , the inclusion map $\iota_S : S \hookrightarrow X$ is continuous.

Corollary

Let X and Y be topological spaces, and suppose $f : X \rightarrow Y$ is continuous.

- i) **RESTRICTING THE DOMAIN:** *The restriction of f to any subspace $S \subseteq X$ is continuous.*
- ii) **RESTRICTING THE CODOMAIN:** *If T is a subspace of Y that contains $f(X)$, then $f : X \rightarrow T$ is continuous.*
- iii) **EXPANDING THE CODOMAIN:** *If Y is a subspace of Z , then $f : X \rightarrow Z$ is continuous.*

Theorem 3.4 (Other Properties of the Subspace Topology)

Suppose S is a subspace of the topological space X .

- i) If $R \subseteq S$ is a subspace of S , then R is a subspace of X ; in other words, the subspace topologies that R inherits from S and from X agree.*
- ii) If $\mathcal{B}$ is a basis for the topology of X , then*

$$\mathcal{B}_S = \{B \cap S : B \in \mathcal{B}\}$$

is a basis for the topology of S .

- iii) if (p_i) is a sequence of points in S and $p \in S$, then $p_i \rightarrow p$ in S if and only if $p_i \rightarrow p$ in X .*
- iv) Every subspace of a Hausdorff space is Hausdorff.*
- v) Every subspace of a first countable space is first countable.*
- vi) Every subspace of a second countable space is second countable.*

Definition

*An injective continuous map that is a homeomorphism onto its image (in the subspace topology) is called a **topological embedding** (or just an **embedding**). If $f : A \rightarrow X$ is such a map, we can think of the image set $f(A)$ as a homeomorphic copy of A inside X .*

Theorem 3.5

Let X be a topological space and let S be a subspace of X . Then the inclusion map $S \hookrightarrow X$ is a topological embedding.

Example 3.6

Let $F : \mathbb{R} \rightarrow \mathbb{R}^2$ be the injective continuous map $F(s) = (s, s^2)$. Its image is the parabola P defined by the equation $y = x^2$. Show that F is a topological embedding.

Proposition 3.7

A continuous injective map that is either open or closed is a topological embedding.

Proposition 3.8

A surjective topological embedding is a homeomorphism.

Example 3.9

If $U \subseteq \mathbb{R}^n$ is open and $f : U \rightarrow \mathbb{R}^k$ is any continuous map, the **graph of f** is the subspace $\Gamma(f) \subseteq \mathbb{R}^{n+k}$ defined by

$$\Gamma(f) = \{(x, f(x)) : x \in U\}.$$

Show that $\Gamma(f)$ is homeomorphic to U by showing that $\Phi : U \rightarrow \mathbb{R}^{n+k}$ defined by $\Phi(x) = (x, f(x))$ is a topological embedding.

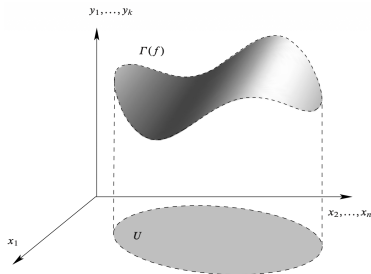


Figure 2: The graph of a continuous function

Example 3.10

Let $N = (0, \dots, 0, 1)$ denote the “north pole” of $\mathbb{S}^n$, and define the **stereographic projection** to be the map $\sigma : \mathbb{S}^n \setminus \{N\} \rightarrow \mathbb{R}^n$ that sends a point $x \in \mathbb{S}^n \setminus \{N\}$ to the point $u \in \mathbb{R}^n$ chosen so that $U = (u, 0)$ is the point in $\mathbb{R}^{n+1}$ where the line through N and x meets the subspace where $x_{n+1} = 0$ (see Figure below) and thus show that $\mathbb{S}^n$ is a topological manifold of dimension n .

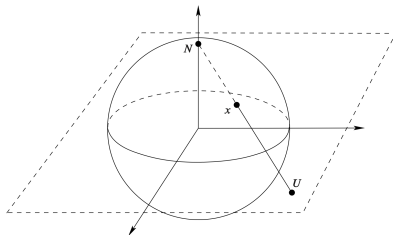


Figure 3: Stereographic projection

Example 3.11

Show that the **doughnut surface**, which is the surface of revolution $D \subseteq \mathbb{R}^3$ obtained by starting with the circle $(x - 2)^2 + z^2 = 1$ in the xz -plane (called the **generating circle**), and revolving it around the z -axis (see Figure below) is 2-manifold. This surface can be parametrised by two angles θ and φ .

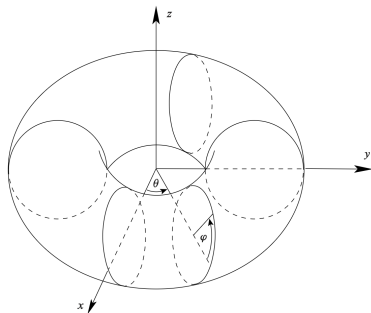


Figure 4: A doughnut surface of revolution

Lemma 3.12 (Gluing Lemma)

Let X and Y be topological spaces, and let A_i be either an arbitrary open cover of X or a finite closed cover of X . Suppose that we are given continuous maps $f_i : A_i \rightarrow Y$ that agree on overlaps:

$$f_i|_{A_i \cap A_j} = f_j|_{A_i \cap A_j}.$$

Then there exists a unique continuous map $f : X \rightarrow Y$ whose restriction to each A_i is equal to f_i .

Theorem 3.13 (Uniqueness of the Subspace Topology)

Suppose S is a subset of a topological space X . The subspace topology on S is the unique topology for which the characteristic property holds.

Proposition 3.14

Suppose $\{(X_j, \mathcal{T}_j)\}_{j=1}^n$ are arbitrary topological spaces. Then the set

$$\mathcal{B} = \left\{ \prod_{k=1}^n U_k : U_k \in \mathcal{T}_k \right\}$$

is a basis for a topology.

Definition

Suppose $\{(X_j, \mathcal{T}_j)\}_{j=1}^n$ are arbitrary topological spaces. On their Cartesian product, we define the **product topology** $\mathcal{T}$ to be the topology generated by the following basis:

$$\mathcal{B} = \left\{ \prod_{k=1}^n U_k : U_k \in \mathcal{T}_k \right\}.$$

The space $\left(\prod_{i=1}^n X_i, \mathcal{T} \right)$ is called a **product space** and the basis subsets of the form $\prod_{k=1}^n U_k, U_k \in \mathcal{T}_k$ are called **product open subsets**. We sometimes write

$$\mathcal{T} = \mathcal{T}_1 \otimes \cdots \otimes \mathcal{T}_n = \bigotimes_{j=1}^n \mathcal{T}_j$$

to explicitly denote the topologies that form the product topology.

Theorem 3.15 (Characteristic Property of the Product Topology)

Suppose $\prod_{j=1}^n X_j$ is a product space. For any topological space Y , a map $f : Y \rightarrow \prod_{j=1}^n X_j$ is continuous if and only if each of its component functions $f_i = \pi_i \circ f$ is continuous, where $\pi^i : \prod_{j=1}^n X_j \rightarrow X_i$ is the canonical projection:

$$\begin{array}{ccc} & X_1 \times \cdots \times X_n & \\ & \downarrow \pi_i & \\ Y & \xrightarrow{f_i} & X_i \end{array}$$

(Note: In the original image, the arrow from Y to $X_1 \times \cdots \times X_n$ is labeled f , and the arrow from Y to X_i is labeled f_i .)

Corollary

If $X_1, \dots, X_n$ are topological spaces, then each canonical projection

$$\pi^i : \prod_{j=1}^n X_j \rightarrow X_i$$

is continuous.

Theorem 3.16 (Uniqueness of the Product Topology)

Let $\{X_i\}_{i=1}^n$ be topological spaces. Then the product topology on their Cartesian product is the unique topology that satisfies the characteristic property.

Theorem 3.17 (Other Properties of the Product Topology)

Let $X_1, \dots, X_n$ be topological spaces.

- i) The product topology is associative, meaning that $\mathcal{T}_1 \otimes \mathcal{T}_2 \otimes \mathcal{T}_3 = \mathcal{T}_1 \otimes (\mathcal{T}_2 \otimes \mathcal{T}_3) = (\mathcal{T}_1 \otimes \mathcal{T}_2) \otimes \mathcal{T}_3$.
- ii) For any $i \in \{1, \dots, n\}$ and any points $x_j \in X_j$, $j \neq i$, the map $f : X_i \rightarrow \prod_{j=1}^n X_j$ given by

$$f(x) = (x_1, \dots, x_{i-1}, x, x_{i+1}, \dots, x_n)$$

is a topological embedding of X_i into the product space.

- iii) Each canonical projection $\pi_i : \prod_{j=1}^k X_j \rightarrow X_i$ is an open map.

iv) If for each i , $\mathcal{B}_i$ is a basis for the topology of X_i , then the set

$$\left\{ \prod_{j=1}^n B_j : B_j \in \mathcal{B}_j \right\}$$

is a basis for the product topology on $\prod_{j=1}^n X_j$.

v) Let S_i is a subspace of X_i for $i = 1, \dots, n$, and let $S = \prod_{j=1}^n S_j$. Then

$$\bigotimes_{j=1}^n \mathcal{T}_{S_j} = \mathcal{T}_S$$

where $\mathcal{T}_S, \mathcal{T}_{S_j}$ are the subspace topologies of S, S_j , respectively.

vi) If each X_i is Hausdorff, so is $\prod_{j=1}^n X_j$.

vii) If each X_i is first countable, so is $\prod_{j=1}^n X_j$.

viii) If each X_i is second countable, so is $\prod_{j=1}^n X_j$.

Definition

Let $f_i : X_i \rightarrow Y_i$ be maps for $i = 1, \dots, k$. Then we define their **product map** $f_1 \times \dots \times f_k : \prod_{j=1}^k X_j \rightarrow \prod_{j=1}^k Y_j$ by

$$f_1 \times \dots \times f_k(x^1, \dots, x^n) = (f_1(x^1), \dots, f_n(x^n)).$$

Proposition 3.18

A product of continuous maps is continuous, and a product of homeomorphisms is a homeomorphism.

Proposition 3.19

If $M_1, \dots, M_k$ are manifolds of dimensions $n_1, \dots, n_k$, respectively, then the product space $\prod_{j=1}^n M_j$ is a manifold of dimension $\sum_{j=1}^n n_j$.

Definition

*The product $\mathbb{T}^n = \mathbb{S}^1 \times \cdots \times \mathbb{S}^1$ of n copies of $\mathbb{S}^1$, which is an n -dimensional manifold called the **n -torus**. In particular, the 2-torus is usually just called the **torus** (plural: **tori**).*

Theorem 3.20

The torus is homeomorphic to the doughnut surface.

Definition

We define the **(abstract) disjoint union** of an indexed family $\{X_\alpha\}_{\alpha \in I}$, denoted by $\coprod_{\alpha \in I} X_\alpha$, to be the set

$$\coprod_{\alpha \in I} X_\alpha = \{(x, \alpha) : \alpha \in I \text{ and } x \in X_\alpha\}.$$

For each index α , there is a natural map $\iota_\alpha : X_\alpha \rightarrow \coprod_{\alpha \in I} X_\alpha$, called the **canonical injection of X_α** defined by $\iota_\alpha(x) = (x, \alpha)$.

Definition

Let $\{X_\alpha\}_{\alpha \in I}$ be an indexed family of topological spaces. We define the **disjoint union topology** on $\coprod_{\alpha \in I} X_\alpha$ by declaring a subset of the disjoint union to be open if and only if its intersection with each set X_α is open in X_α . With this topology, $\coprod_{\alpha \in I} X_\alpha$ is called a **disjoint union space**.

Theorem 3.21 (Characteristic Property of Disjoint Union Spaces)

Suppose that $\{X_\alpha\}$ is an indexed family of topological spaces, and Y is any topological space. A map $f : \coprod_{\alpha \in I} X_\alpha \rightarrow Y$ is continuous if and only if its restriction to each X_α is continuous. The disjoint union topology is the unique topology on $\coprod_{\alpha \in I} X_\alpha$ with this property.

Theorem 3.22 (Other Properties of Disjoint Union Spaces)

Let $\{X_\alpha\}_{\alpha \in I}$ be an indexed family of topological spaces.

- i) A subset of $\coprod_{\alpha \in I} X_\alpha$ is closed if and only if its intersection with each X_α is closed.
- ii) Each canonical injection $\iota_\sigma : X_\sigma \rightarrow \coprod_{\alpha \in I} X_\alpha$ is a topological embedding and an open and closed map.
- iii) If each X_α is Hausdorff, then so is $\coprod_{\alpha \in I} X_\alpha$.
- iv) If each X_α is first countable, then so is $\coprod_{\alpha \in I} X_\alpha$.
- v) If each X_α is first countable and the index set I is countable, then $\coprod_{\alpha \in I} X_\alpha$ is second countable.

Definition

Let X be a topological space, Y any set, and $q : X \rightarrow Y$ be a surjective map. Define the topology on Y by declaring a subset $U \subseteq Y$ to be open if and only if $q^{-1}(U)$ is open in X . This is called the **quotient topology** induced by the map q .

Definition

*If X and Y are topological spaces, a map $q : X \rightarrow Y$ is called a **quotient map** if it is surjective and Y has the quotient topology induced by q .*

Definition

Let $\sim$ be an equivalence relation on a topological space X . For each $p \in X$, let $[p]$ denote the equivalence class of p , and let $X/\sim$ be the quotient set. Let $q : X \rightarrow X/\sim$ be the natural projection sending each element of X to its equivalence class. Then q is surjective and $X/\sim$ together with the quotient topology induced by q is called the **quotient space** (or sometimes **identification space**) of X by the given equivalence relation.

Example 3.23

Let X be the unit square in $\mathbb{R}^2$. Consider the equivalence relation generated by

$$(0, t) \sim (1, 1 - t).$$

The quotient space is called the **Möbius strip**.

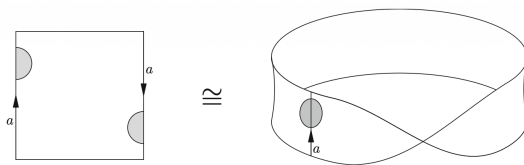


Figure 5: A Möbius strip

Theorem 3.24

Suppose P is a second countable space and M is a quotient space of P . If M is locally Euclidean, then it is second countable.

Theorem 3.25

Suppose $q : X \rightarrow Y$ is an open quotient map. Then Y is Hausdorff if and only if the set $\mathcal{R} = \{(x_1, x_2) : q(x_1) = q(x_2)\}$ is closed in $X \times X$.

Corollary

Suppose $\sim$ is an equivalence relation on a space X . If the quotient map $X \rightarrow X/\sim$ is an open map, then $X/\sim$ is Hausdorff if and only if $\sim$ is closed subset of $X \times X$.

Definition

Suppose $q : X \rightarrow Y$ is a map. Any subset of the form $q^{-1}(y) \subseteq X$ for some $y \in Y$ is called a **fiber of q** . A subset $U \subseteq X$ is said to be **saturated with respect to q** if $U = q^{-1}(V)$ for some subset $V \subseteq Y$.

Theorem 3.26

Let $q : X \rightarrow Y$ be a map. For a subset $U \subseteq X$, the following are equivalent.

- i) U is saturated.*
- ii) $U = q^{-1}(q(U))$.*
- iii) U is a union of fibers.*
- iv) If $x \in U$, then every point $x' \in X$ such that $q(x) = q(x')$ is also in U .*

Theorem 3.27

A continuous surjective map $q : X \rightarrow Y$ is a quotient map if and only if it takes saturated open subsets to open subsets, or saturated closed subsets to closed subsets.

Theorem 3.28 (Properties of Quotient Maps)

- i) *Any composition of quotient maps is a quotient map.*
- ii) *An injective quotient map is a homeomorphism.*
- iii) *If $q : X \rightarrow Y$ is a quotient map, a subset $K \subseteq Y$ is closed if and only if $q^{-1}(K)$ is closed in X .*
- iv) *If $q : X \rightarrow Y$ is a quotient map and $U \subseteq X$ is a saturated open or closed subset, then the restriction $q|_U : U \rightarrow q(U)$ is a quotient map.*
- v) *If $\{q_\alpha : X_\alpha \rightarrow Y_\alpha\}$ is an indexed family of quotient maps, then the map $q : \coprod_\alpha X_\alpha \rightarrow \coprod_\alpha Y_\alpha$ whose restriction to each X_α is equal to q_α is a quotient map.*

Theorem 3.29

If $q : X \rightarrow Y$ is a surjective continuous map that is also an open or closed map, then it is a quotient map.

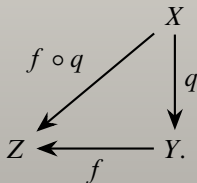
Theorem 3.30

Suppose X and Y are topological spaces, and $f : X \rightarrow Y$ is a continuous map that is either open or closed.

- i) If f is injective, it is a topological embedding.*
- ii) If f is surjective, it is a quotient map.*
- iii) If f is bijective, it is a homeomorphism.*

Theorem 3.31 (Characteristic Property of the Quotient Topology)

Suppose X and Y are topological spaces and $q : X \rightarrow Y$ is a quotient map. For any topological space Z , a map $f : Y \rightarrow Z$ is continuous if and only if the composite map $f \circ q$ is continuous:



Theorem 3.32 (Uniqueness of the Quotient Topology)

Given a topological space X , a set Y , and a surjective map $q : X \rightarrow Y$, the quotient topology is the only topology on Y for which the characteristic property holds.

Definition

A **topological group** is a group $(G, \star)$ endowed with a topology such that the maps $\star : G \times G \rightarrow G$ and $i : G \rightarrow G$ given by

$$\star(g_1, g_2) = g_1 \star g_2, \quad i(g) = g^{-1}$$

are continuous (continuity of $\star$ is understood to be with respect to the product topology on $G \times G$).

Example 3.33

Each of the following is a topological group:

- *the real line $\mathbb{R}$ with its additive group structure and Euclidean topology;*
- *the set $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$ of nonzero complex numbers under multiplication, with the Euclidean topology;*
- *the general linear group $GL(n, \mathbb{C})$, the set of $n \times n$ invertible complex matrices under matrix multiplication, with the subspace topology obtained from $\mathbb{R}^{2n^2}$ (where we identify an $n \times n$ matrix with a point in $\mathbb{R}^{2n^2}$ by using the matrix entries as coordinates);*
- *any group whatsoever, with the discrete topology (any such group is called a **discrete group**).*

Theorem 3.34

Any subgroup of a topological group is a topological group with the subspace topology. Any finite product of topological groups is a topological group with the direct product group structure and the product topology.

Example 3.35

In view of the previous theorem, each of the following is a topological group, with the product topology or subspace topology as appropriate:

- *Euclidean space $\mathbb{R}^n = \mathbb{R} \times \cdots \times \mathbb{R}$ as a group under vector addition;*
- *the circle $\mathbb{S}^1 \subseteq \mathbb{C}^*$ under complex multiplication;*
- *the n -torus $\mathbb{T}^n = \mathbb{S}^1 \times \cdots \times \mathbb{S}^1$;*
- *the orthogonal group $O(n)$.*

Definition

If G is a topological group and $g \in G$, **left translation by g** is the homeomorphism $L_g : G \rightarrow G$ defined by $L_g(g') = gg'$. Similarly, **right translation by g** is the homeomorphism defined by $R_g(g') = g'g$.

Definition

A topological space X is said to be **topologically homogeneous** if for any $x, y \in X$, there is a homeomorphism $\varphi : X \rightarrow X$ taking x to y .

Corollary

Every topological group is topologically homogeneous.

Definition

Suppose G is a group, and X is a set. A **left action of G on X** is a map $G \times X \rightarrow X$, written $(g, x) \rightarrow g \cdot x$, with the following properties:

- i) $g_1 \cdot (g_2 \cdot x) = (g_1 g_2) \cdot x$ for all $x \in X$ and all $g_1, g_2 \in G$.
- ii) $1 \cdot x = x$ for all $x \in X$.

Similarly, a **right action** is a map $X \times G \rightarrow X$, written $(x, g) \rightarrow x \cdot g$, with the same properties except that composition works in reverse: $(x \cdot g_1) \cdot g_2 = x \cdot (g_1 g_2)$. Any right action determines a left action in a canonical way, and vice versa, by the correspondence

$$g \cdot x = x \cdot g^{-1}.$$

Definition

Let X be a topological space and let G be a group acting on the left of X . The action is called an **action by homeomorphisms** if for each $g \in G$, the map $x \mapsto g \cdot x$ is a homeomorphism of X . If in addition G is a topological group, the action is said to be **continuous** if the map $G \times X \rightarrow X$ is continuous. The next theorem explains the relationship between the two concepts.

Theorem 3.36

Suppose G is a topological group acting on a topological space X .

- i) If the action is continuous, then it is an action by homeomorphisms.*
- ii) If G has the discrete topology, then the action is continuous if and only if it is an action by homeomorphisms.*

Definition

For any $x \in X$, the set $Gx = \{g \cdot x : g \in G\} \subseteq X$ is called the **orbit of x** . The action is said to be **transitive** if for every pair of points $x, y \in X$, there is a group element g such that $g \cdot x = y$, or equivalently if $Gx = X$ for each $x \in X$. The **stabilizer** of x is $G_x = \{g \in G : g \cdot x = x\}$. The action is said to be **free** if the only element of G that fixes any point in X is the identity; that is, if $G_x = \{e\}$ for each x . The action is called **effective** if $g \cdot x = x$ for all x implies that $g = e$.

Example 3.37

- *The general linear group $GL(n, \mathbb{R})$ acts on the left on $\mathbb{R}^n$ by matrix multiplication, considering each vector in $\mathbb{R}^n$ as a column matrix. Given any nonzero vector $x \in \mathbb{R}^n$ we can find vectors $x_2, \dots, x_n$ such that $\{x, x_2, \dots, x_n\}$ is a basis for $\mathbb{R}^n$, and then the matrix g with columns $(x, x_2, \dots, x_n)$ is invertible and takes the vector $(1, 0, \dots, 0)$ to x . If y is any other nonzero vector, the same argument shows that there is a matrix $h \in GL(n, \mathbb{R})$ taking $(1, 0, \dots, 0)$ to y , and then hg^{-1} takes x to y . Thus there are only two orbits: $\mathbb{R}^n \setminus \{0\}$ and $\{0\}$.*
- *The group $\mathbb{R}^*$ acts on $\mathbb{R}^n \setminus \{0\}$ by scalar multiplication. The action is free, and the orbits are the lines through the origin (with the origin removed).*
- *Any topological group G acts continuously*

Example 3.38

- Any topological group G acts continuously, freely and transitively on itself on the left by left translation: $g \cdot g' = L_g(g') = gg'$. Similarly, G acts on itself on the right by right translation.
- If Γ is a subgroup of the topological group G (with the subspace topology), then the group multiplication on the left or right defines a left or right action of Γ on G ; it is just the restriction of the action of G on itself to $\Gamma \times G$ or $G \times \Gamma$. This action is still continuous and free but in general, not transitive.
- The two-element discrete group $\mathbb{S}^0$ acts freely on $\mathbb{S}^n$ by multiplication: $\pm 1 \cdot x = \pm x$. This is an action by homeomorphisms, and because the group is discrete, it is continuous. Each orbit is a pair of antipodal points: $\{x, -x\}$.

Definition

*Given an action of a group G on a space X , we define the relation on X by saying $x_1 \sim x_2$ if there is an element $g \in G$ such that $g \cdot x_1 = x_2$. This is an equivalence relation and the equivalence classes are precisely the orbits of the group action. The resulting quotient space is denoted by X/G , and is called the **orbit space** of the action. If the action is transitive, the orbit space is a single point, so only nontransitive actions yield interesting examples.*

Example 3.39

- *The quotient space of $\mathbb{R}^n$ by $O(n)$ is homeomorphic to $[0, \infty)$.*
- *The real projective space $\mathbb{P}^n$ is the orbit space of the action of $\mathbb{R}^*$ on $\mathbb{R}^{n+1} \setminus \{0\}$ by scalar multiplication.*

Definition

*Let G be a topological group and consider the action of a subgroup Γ on G . An orbit of the right action of Γ on G is a set of the form $\{g\gamma : \gamma \in \Gamma\}$, which is precisely the left coset $g\Gamma$. Thus the orbit space of the right of Γ on G is the set G/Γ of left cosets with the quotient topology. This quotient space is called the **(left) coset space** of G by Γ .*

Example 3.40

Consider the coset space $\mathbb{R}/\mathbb{Z}$. Because $\mathbb{Z}$ is a subgroup of $\mathbb{R}$, there is a natural free continuous action of $\mathbb{Z}$ on $\mathbb{R}$ by translation $n \cdot x = n + x$. Consider also the **exponential quotient map** $\varepsilon : \mathbb{R} \rightarrow \mathbb{S}^1$ defined by

$$\varepsilon(r) = e^{2\pi i r}.$$

It is straightforward to check that this is a local homeomorphism and thus an open map, so it is a quotient map. Because it makes the same identification as the quotient map $\mathbb{R} \rightarrow \mathbb{R}/\mathbb{Z}$, the uniqueness of quotient spaces tells us that $\mathbb{R}/\mathbb{Z}$ is homeomorphic to $\mathbb{S}^1$.